Perceiving Spectral Variation: Unsupervised Spectrum Motion Feature Learning for Hyperspectral Image Classification

Yifan Sun, Bing Liu, Xuchu Yu, Anzhu Yu, Kuiliang Gao, and Lei Ding

***Abstract*— In recent years, deep-learning-based hyperspectral image (HSI) classification methods have achieved significant development. The superior capability of feature extraction from these data-driven methods dramatically improves classification performance. However, the previous methods usually require retraining the network from scratch to obtain the capability of feature extraction adaptive for the target image when facing a new HSI to be classified, which is a time-consuming and redundant process. In this article, we consider putting this process ahead and making the network have a robust capability of feature extraction with generalization through pretraining. Therefore, the network enables to directly extract features of the target HSI without retraining. For this purpose, we rethink the 3-D HSI data from the perspective of spectral sequence, and we attempt to extract the spectral variation information as the spectrum motion feature. Then, we construct an unsupervised spectrum motion feature learning framework (SMF-UL), which can be pretrained on mass unlabeled HSI data to learn the knowledge about perceiving spectral variation. Furthermore, to achieve the expansion of source data for pretraining, we develop an extendable training dataset construction method, which can integrate HSIs of different sizes, the number of bands, and sensors into a unified training set to utilize the rapidly growing mass unlabeled HSI data effectively. Finally, we use the trained network to directly extract the spectrum motion feature of the target HSI for classification, so the laborious retraining of the network can be avoided. Extensive experiments show that the proposed SMF-UL acquires the robust capability of feature extraction with generalization through unsupervised learning on mass unlabeled HSI data, and the classification performance of extracted spectrum motion feature is competitive to advanced in-domain and cross-domain methods, which shows its flexibility and superiority. The code of SMF-UL will be open at https://github.com/sssssyf/SMF-UL.**

***Index Terms*— Feature extraction, hyperspectral image (HSI), motion feature, optical flow, unsupervised learning.**

I. Introduction

HYPERSPECTRAL remote sensing technology has fully released the vast potential in geospatial information observation because of its unique technical characteristics in mass data analysis, information fusion, and fine recognition of ground object elements [1]. The capacity of hyperspectral image (HSI) processing and analysis will directly determine the level of ground object recognition [2]. HSI classification technology is the basic approach to achieve the recognition of ground object attributes, which involves labeling categories of ground objects pixel-by-pixel.

To effectively utilize abundant spectral information in HSI and alleviate the contradictions of high-dimension nonlinearity, spectral correlation, spectral mixing, and so on, feature extraction is usually utilized as a vital process for classification. Feature extraction focuses on a transformation of the original HSI feature to better distinguish different ground objects [3]. Early shallow feature extraction methods include some handcrafted feature designs as guidelines and some other subspace methods based on a statistic or learning [4], [5], [6], [7], [8], [9], [10], [11]. Although these works make a significant contribution to the hyperspectral community, such complicated and shallow handcraft features greatly rely on expertise, and it is hard to adapt to various hyperspectral scenes.

The rise of deep learning enables full automation of feature extraction with data-driven to be a reality. Deep-learning-based HSI classification methods could implicitly perform automatic feature extraction by constructing an end-to-end framework that integrates feature extraction and classification process, which are capable of avoiding laborious specialized feature design and obtaining deep and robust feature conducive for classification. Therefore, the development of classification methods based on deep learning is closely linked with the level of feature extraction, which is also approximately similar to the evolution history of traditional feature extraction methods.

The spectral-feature-based frameworks that consider each spectrum of an HSI pixel as a unit set are initially introduced [12], [13], [14], [15]. Then, numerous studies concentrate on how to effectively combine the spectral and spatial feature [16], [17]. Among them, the convolutional neural network (CNN) plays a prominent role in spatial feature extraction. Many subsequent works have focused on exploring and designing networks with superior performance to efficiently extract feature conducive to classification. To jointly extract spectral–spatial features, the backbone based on 3-D-CNN gradually takes a dominant position [18], [19]. Other novel backbones, such as capsule network [20], long short-term

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memory (LSTM) [21], and transformer [22], are introduced to further learn the joint feature representation. Besides, the attention mechanism (AM) is widely utilized to enhance feature representation in the networks [23], [24], [25].

The abovementioned works usually take data patches as input, so the long-distance contextual dependence of the feature is hard to capture. Thus, some researchers introduce the fully convolutional neural network (FCN) as the backbone to extract the global feature [26], [27], [28], [29], [30], which performs well on accuracy and efficiency. The graph convolutional network (GCN) [31], [32], [33] is introduced to further break the limitation results from the convolution kernel of fixed size. These methods have practically shown the superb performance of feature extraction and classification under the condition of adequate annotated samples. To mitigate the subsistent difficulty of the scarcity of annotated samples and the laborious process of annotating, numerous studies based on unsupervised learning [34], [35], [36], [37], self-supervised learning [38], [39], [40], semisupervised learning [41], [42], and so on are explored to enhance the feature extraction without adequate annotated samples.

However, a common characteristic of the current prevalent works mentioned above is that they require retraining the network from scratch to obtain the feature extraction capability adaptive for the target HSI to be classified. This is a time-consuming and redundant process in practice, which cannot meet the requirements of rapidly processing and analyzing massive hyperspectral data. Aiming at this practical problem, we want to raise two questions: 1) can the feature learning process be taken ahead on the source dataset without annotated samples so that the feature extractor can maintain the robust capability of feature extraction with generalization when facing arbitrary new HSIs? and 2) can the rapidly growing mass of unlabeled HSI data be constructed into a unified and extensible source dataset to bridge the gap caused by different sensors, arbitrary size of images, diverse number of bands, and discrepant spectral range?

To solve the two questions, how to make the feature extractor have the capability of feature extraction with generalization through learning from different HSI data is a primary point. Motivated by the several works that consider the spectral sequence information [14], [15], [21], we also look at hyperspectral data in terms of spectral sequences. As shown in Fig. 1, any spectral sequence has its variation characteristics, that is, there is a motion vector $\boldsymbol{F}$ at any point of the spectral curve. $\boldsymbol{F}$ denotes the direction and degree of variation at the current point of the spectral curve, which can be divided into a horizontal component $\boldsymbol{u}$ and a vertical component $\boldsymbol{v}$. This also means that the ground objects of the same objects have similar spectrum variation characteristics, while the different ones have the discrepant variation trend of the spectrum. Thus, the motion feature brought about by spectral variation is a beneficial supplement to the original feature. Such global spectral variation reflected in the HSI is the brightness changes of single-band gray images.

Optical flow estimation is usually utilized to capture the motion information of moving objects motion information in the computer vision (CV) field [43], [44], [45]. Except

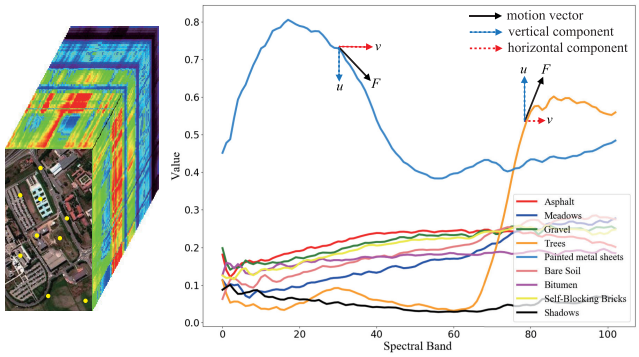


Fig. 1. Representation of spectral variation and spectrum motion vector on the HSI data. (The PU that contains nine categories of ground objects is taken as an example, and one spectral sequence is selected for each category.)

for the optical flow generated by the moving objects, the brightness change of the static scene also causes the optical flow. Therefore, we consider that the optical flow estimation can be also used to extract the brightness change on the HSI with the static scene without moving objects.

Then, we introduce the specific photometric loss function [46], [47] and a lightweight PWC-Net [48] from the optical flow estimation task to construct our unsupervised spectrum motion feature learning framework (SMF-UL). Furthermore, we construct a training dataset for the learning framework by extracting three consecutive bands from HSI as a three-channel image. This construction strategy and the same cropped size of different HSIs guarantee that the training dataset is extensible for any HSI data with different specification parameters. It also means the arbitrary HSI data can be used for feature learning, which enables solving the problem of insufficient training samples and effectively utilizing the rapidly growing unlabeled HSI data simultaneously. Besides, the SMF-UL will acquire a more robust capability of feature extraction with generalization when pretraining on a large-size source dataset. The main contributions of our work can be summarized as follows.

1) Aiming at the characteristics of spectral variation, we construct an SMF-UL to learn the spectral variation knowledge on HSI data for classification. A joint unsupervised strategy of multiresolution features is proposed to utilize the multiple outputs of the lightweight PWC-Net [48] to jointly calculate the unsupervised loss function.
2) Aiming at the laborious retraining of the currently prevalent methods, we propose to take ahead the feature learning process, and the pretraining makes that the feature extractor with generalization capability can be directly used to extract spectrum motion feature of the target HSI to be classified without retraining.
3) Aiming at achieving the expansion of source data and effectively utilizing the massive unlabeled HSI data, we develop an extendable training dataset construction method, which can integrate arbitrary HSI data with different specification parameters (such as sensor, image size, and the number of bands) into a unified training dataset.

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4) Extensive experiments about in-domain and cross-domain on 11 groups of open-source HSI data (the four groups are the target data to be classified) show that the proposed method is capable of effectively combining the efficiency and performance through pretraining compared with currently advanced methods, which shows its flexibility and superiority.

Section II describes the related works. Section III describes the SMF-UL. Section IV is experimental verification and analysis. Section V concludes this article comprehensively.

II. Related Works

The deep-learning-based HSI classification methods can be roughly divided into three learning paradigms according to the condition of utilizing the labeled and unlabeled samples: supervised feature learning, semisupervised feature learning, and unsupervised feature learning.

A. Supervised Feature Learning

Supervised feature learning refers to the whole participation of labeled samples as supervised signals during feature learning. Most of the prevalent end-to-end classification frameworks belong to supervised feature learning, such as spectral feature learning methods represented by 1D-CNN [13], SpectralFormer [15], and so on, and joint-spectral–spatial feature learning methods represented by the influential SSRN [18], DBDM [23], SSTN [22], and so on. Aiming at the classification of the small samples, DFSL [49], RN-FSC [50], and In-Net [51] pretrain the network on a source dataset with supervised learning, so the network can adapt to the target dataset under the condition of small samples. Such a similar training schedule is also common in some methods based on transfer learning [52], [53], [54], [55], and their networks are usually required to fine-tune to further adapt to the target domain. As an improved alternative, our method performs unsupervised feature learning on unlabeled samples, so avoiding the dependence on labeled samples. Fine-tuning is optional and even can be discarded in practice, which profits from the learned ability of feature extraction with generalization. Besides, DCFSL [56] and Gia-CFSL [57] explore a few-shot learning framework jointly trained by all labeled samples of source data and a few labeled samples of target data to further improve the performance of small samples classification.

B. Semisupervised Feature Learning

Because of the laborious annotation of labels with expertise knowledge, HSIs usually merely have limited labeled samples but also accompany abundant unlabeled samples. Semisupervised feature learning can combine unlabeled samples and labeled samples for joint learning, hence learning a more robust discriminant feature. As a representative semisupervised method, the generative adversarial network (GAN) utilizes the generator to generate pseudolabels. It then sends labeled and pseudolabeled samples into the discriminator to implicitly achieve the semisupervised feature learning by modeling the high-dimensional distribution of data [58], [59], [60]. The other is GCN that considers both the features of adjacent graph nodes and the adjacent spatial pixels, so the information of labeled and unlabeled samples can be modeled simultaneously [31], [32], [33]. Besides, the pseudolabel-based methods usually utilize unlabeled samples to expand the number of training data, so alleviating the dependence on labeled samples when feature learning [41], [61]. 3DAES utilizes an autoencoder to learn features with unlabeled data and constructs a Siamese network to learn supervised features on limited labeled samples to rectify unsupervised features [42]. These works with semisupervised learning show that the participation of the number of unlabeled samples is conducive to learning more robust features when labeled samples are scarce.

C. Unsupervised Feature Learning

Unsupervised feature learning refers to the automatic learning of feature representation from the data without any annotated information. It raises as a solution for learning features from mass HSI unlabeled data. An encoder–decoder architecture is utilized to reconstruct the input patterns to achieve the spatial–spectral feature learning with unlabeled HSI data [34], [35]. The unsupervised structure knowledge learning on unlabeled HSI data is helpful to improve the supervised classification [36]. UM^{2}L [37] is the first attempt at unsupervised metalearning, which constructs multiview features on unlabeled HSI data to train the relation network, so the dependence on labeled samples in supervised metalearning tasks can be avoided. Besides, as a significant branch of unsupervised feature learning, self-supervised feature learning enables to extract data representation as the supervised signal through designing the proxy tasks. The pivotal issue of self-supervised feature learning is how to construct an unsupervised loss function by using some attributes of data instead of labeled samples. It involves contrastive learning and generative learning, and feature learning based on contrastive learning for HSI classification has received much attention. Construction of the special contrastive loss function brings samples with the same category (positive samples) closer and extends the distance between samples with different categories (negative samples) in the high-dimension feature space [38], [39], [40]. A marginalized graph self-representation (MGSR) method is proposed to perform unsupervised band selection by extracting the spatial structural information with super pixel segmentation and marginalizing out the corrupted pixels [62]. A multiview fusion clustering framework (MVEC) based on an extreme learning machine (ELM) is developed to unsupervised learn the embedding from each view of the data and automatically weight the contribution according to the correlations and dependencies between each [63]. In the above works, except for a cross-domain version provided by UM^{2}L, other methods need to retrain the network for the target image (i.e., in-domain). Therefore, the UM^{2}L is the work most relevant to us, but its network still needs a relatively time-consuming pretraining and fine-tuning process because of its end-to-end relation network and patchwise input form. Besides, it also accompanies some classification map problems about the expansion of ground objects' boundary

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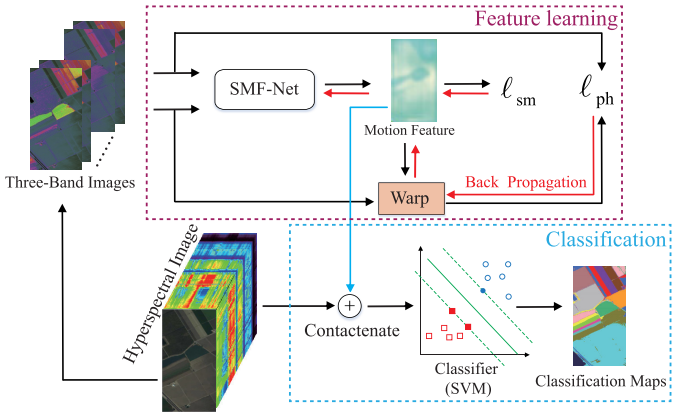


Fig. 2. Processing flow of the proposed unsupervised spectrum motion feature learning method for HSI classification.

and distortion of ground objects' distribution. In contrast, the proposed SMF-UL enables to achieve improvements on these points, which will be analyzed subsequently in detail.

III. Proposed Method

The processing flow of the proposed classification method can be divided into such stages: unsupervised spectrum motion feature learning (pretraining), fine-tuning (optional), feature extraction, and classification.

A. Unsupervised Spectrum Motion Feature Learning

The in-domain version is illustrated first in this section for conciseness, as shown in Fig. 2. Given HSI data, $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote the height, the width, and the number of bands, respectively. First, aiming at the characteristics of spectral variation, the purpose of our SMF-UL is to learn the motion feature without any labels. Because the global spectral variation is reflected in the HSI in the brightness changes of single-band gray images, we consider that the optical flow estimation can be also used to extract the brightness change on the HSI with the static scene without moving objects. For this purpose, we introduce the optical flow network as the basis of our spectrum motion feature network (SMF-Net). Therefore, the inputs should be two adjacent bands $\boldsymbol{B}_i, \boldsymbol{B}_{i+1} \in \mathbb{R}^{H\times W}(i = 1, 2, \ldots, C-2)$, so the spectral variation between them can be captured. Since the input requirements are RGB images, so we take three consecutive single-band images in HSI as a three-channel image, and its spectral values are normalized in the range of (0, 255). Therefore, the spectral difference of two adjacent bands $\boldsymbol{B}_i, \boldsymbol{B}_{i+1}$ is transformed into the brightness variation information of two three-channel images $\boldsymbol{I}_i(\boldsymbol{B}_{i-1}, \boldsymbol{B}_i, \boldsymbol{B}_{i+1})$, $\boldsymbol{I}_{i+1}(\boldsymbol{B}_i, \boldsymbol{B}_{i+1}, \boldsymbol{B}_{i+2})$.

If we take $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$ as an example, the goal of unsupervised learning is to train the SMF-Net $\psi(\cdot)$ to extract the spectrum motion feature $\boldsymbol{F} = \{\boldsymbol{u}, \boldsymbol{v}\} \in \mathbb{R}^{H\times W\times 2}$ for two consecutive three-channel images $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$

$$\boldsymbol{F} = \psi(\boldsymbol{I}_1, \boldsymbol{I}_2; \Theta) \tag{1}$$

where Θ is the learnable parameters of the network. To construct the unsupervised motion feature learning task, the spectral values of $\boldsymbol{I}_2$ are transformed to align with $\boldsymbol{I}_1$ with the spectrum motion feature $\boldsymbol{F}$ through warping

$$\hat{\boldsymbol{I}}_1(p) = \boldsymbol{I}_2(p + \boldsymbol{F}(p)) \tag{2}$$

where p represents the same location of pixels in two images. Since the HSI is a static scene with no pixels occlusion by moving objects, we can directly utilize the popular photometric loss function [46] to construct our motion feature learning task, which can be formulated as

$$\ell_{\text{ph}} \sim \sum_p \rho(\hat{\boldsymbol{I}}_1(\Theta), \boldsymbol{I}_1) \tag{3}$$

where $\rho(\cdot)$ is the pixelwise similarity measurement, and we utilize the structural similarities (SSIMs) in our learning framework to narrow the spectral value gap between $\boldsymbol{I}_1$ and the transformed image $\hat{\boldsymbol{I}}_1$. Furthermore, the single photometric loss function may lead to an ambiguity in somewhere textureless or with repetitive patterns [47]. Therefore, a common smooth regularization is introduced to mitigate the ambiguity, so as to enhance the representation of motion features, which can be formulated as

$$\ell_{\text{sm}} \sim \sum_{d\in x,y} \sum_p \|\nabla_d \boldsymbol{F}\|_1 e^{-|\nabla_d \boldsymbol{I}_1|}. \tag{4}$$

Thus, the unsupervised loss function is composed of photometric loss ℓ_{ph} and smooth regularization ℓ_{sm}

$$\ell = \ell_{\text{ph}}(\boldsymbol{F}) + \lambda \ell_{\text{sm}}(\boldsymbol{F}) \tag{5}$$

where λ is a contribution weight of the second term. Therefore, by decreasing the total loss ℓ, we train the network by the back-propagation algorithm to update the learnable parameters set Θ. After the unsupervised feature learning, we utilize the trained SMF-Net to extract the spectrum motion feature, which is concatenated with the original HSI as input of the classifier SVM for classification.

B. Spectrum Motion Feature Network

In recent years, the advanced pretrained backbones represented by ResNet are indeed proven superior models to extract features, and they have been applied widely to the feature extraction of HSI because of the powerful deep feature learning and extraction ability [54]. However, the extracted feature of the deep network represented by ResNet is usually a high-dimension deep abstract feature that enables to identify the corresponding category of input in HSI classification, so the HSI is usually required to be cut into local patches as inputs. Such a classification pattern as mainstream has achieved huge development in recent years. However, the core innovation of our method is to learn spectral variation information through the constructed unsupervised learning framework, rather than directly extracting the usual deep high-dimension features utilized to identify. Therefore, the deep network represented by ResNet will not apply, and we utilize the optical flow estimation network PWC-Net to construct our SMF-UL.

In fact, an arbitrary advanced optical flow network enables to be the SMF-Net for our unsupervised learning framework.

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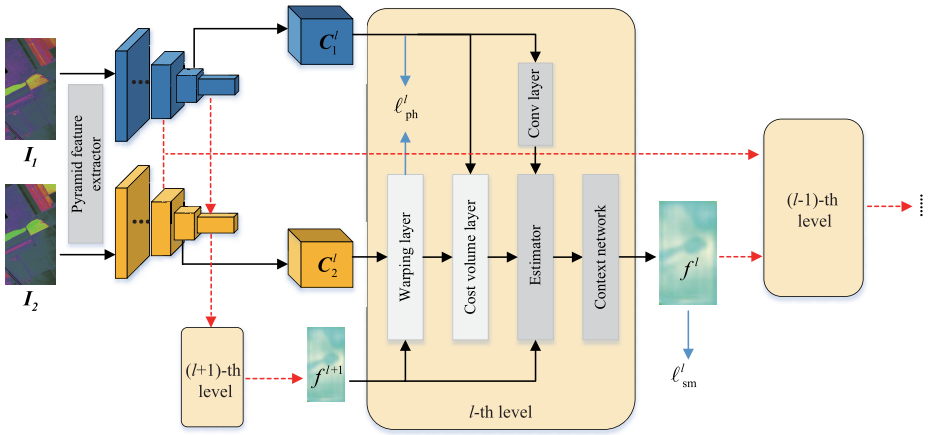


Fig. 3. Illustration of the SMF-Net and the cross-layer loss function calculation based on the lightweight PWC-Net [48]. (In addition to the current lth level, the red dotted arrows indicate the information flow of the previous $(l+1)$th level and subsequent $(l-1)$th level, and the blue arrows indicate the calculation of the loss terms.)

TABLE I
DETAILED SETTING OF LEARNABLE CONVOLUTION PARAMETERS IN THE LIGHTWEIGHT PWC-NET [48]

Layer Name	Level	Dimension	Kernel Size	Stride	Layer Name	Dimension	Kernel Size	Dilation
Pyramid Feature Extractor	0	(3, 16)	3×3	2	Estimator	($\frac{1}{2}$dim$_l$+32+2, 128)	3×3	1
		(16, 16)	3×3	1		(128, 128)	3×3	1
	1	(16, 32)	3×3	2		(128+128, 96)	3×3	1
		(32, 32)	3×3	1		(128+96, 64)	3×3	1
	2	(32, 64)	3×3	2		(96+64, 32)	3×3	1
		(64, 64)	3×3	1	Context Network	(32+2, 128)	3×3	1
	3	(64, 96)	3×3	2		(128, 128)	3×3	2
		(96, 96)	3×3	1		(128, 128)	3×3	4
	4	(96, 128)	3×3	2		(128, 96)	3×3	8
		(128, 128)	3×3	1		(96, 64)	3×3	16
	5	(128, 192)	3×3	2		(64, 32)	3×3	1
		(192, 192)	3×3	1		(32, 2)	3×3	1
Conv Layer	l	(dim$_l$, 32)	1×1	1				

In this article, to make the training easy, we utilize the lightweight PWC-Net [48], and it is a compact version of PWC-Net that performs well in the supervised optical flow estimation [64]. To incorporate it into our unsupervised feature learning framework and effectively utilize its multiresolution outputs, we propose a joint unsupervised strategy of multiresolution features, which utilizes the multiresolution outputs of different levels to commonly calculate the photometric loss ℓ_{ph} and smooth regularization ℓ_{sm}, as shown in Fig. 3. To make the details of PWC-Net clearer, its detailed setting of learnable convolution parameters is shown in Table I. For two three-channel images $\boldsymbol{I_1}, \boldsymbol{I_2} \in \mathbb{R}^{H\times W\times 3}$ in HSI, their pyramid features with L-level are first extracted by a pyramid extractor, which consists of $L-1$ convolution layers with a downsampling factor of 2. The bottom feature is the input, i.e., $\boldsymbol{C}_t^0 = \boldsymbol{I}_t$, so the top feature is $\boldsymbol{C}_t^L$. At the lth level $(1 \le l < L)$, the warping operation can be formulated as

$$\boldsymbol{C}_{\text{warp}}^l(p) = \boldsymbol{C}_2^l(p + \text{Upsample}(\boldsymbol{f}^{l+1})(p)) \tag{6}$$

where $\boldsymbol{f}^{l+1}$ denotes the motion feature calculated by the former-level (i.e., l+1 level) pyramid features, and it should be zero at the top level. Thus, the photometric loss ℓ_{ph} and smooth regularization ℓ_{sm} at the lth level can be formulated as

$$\ell_{\text{ph}}^l \sim \sum_p \rho\left(\boldsymbol{C}_{\text{warp}}^l(p), \boldsymbol{C}_1^l\right) \tag{7}$$

$$\ell_{\text{sm}}^l \sim \sum_{d\in x,y}\sum_p \left\|\nabla_d \boldsymbol{f}^l\right\|_1 e^{-|\nabla_d \boldsymbol{C}_1^l|}. \tag{8}$$

Then, the cost volume that denotes the matching cost between two images at the lth level can be formulated as

$$\textbf{CostV}^l(p) = \frac{1}{N}\left(\boldsymbol{C}_1^l(p)\right)^\top \boldsymbol{C}_{\text{warp}}^l(p) \tag{9}$$

where N is the length of the column vector $\boldsymbol{C}_1^l(p)$. Next, the results of cost volume layer $\textbf{CostV}^l$, $\boldsymbol{C}_1^l(p)$, and $\boldsymbol{f}^{l+1}$ after upsampling are as input of the estimator. The estimator of the lightweight version reduces the dense connections to decrease the number of feature channels. Besides, a context network that consists of multiple dilated convolution layers with increasing dilation coefficients is deployed to refine the feature further. The process is iterative until the bottom level (zeroth level), and the information flowing at the lth level is mainly shown in Fig. 3. Finally, to effectively utilize the

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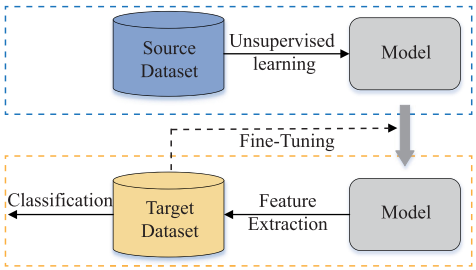


Fig. 4. Illustration for the strategy of cross-domain spectrum motion feature extraction.

multiresolution features of different layers, the photometric loss and smooth regularization of all levels are summed up to the total loss function, so avoiding the information loss resulting from the single loss function, which can be formulated as

$$\ell = \sum_{1\le l\le L}^{l} \ell_{\text{ph}}^{l} + \lambda \sum_{1\le l\le L}^{l} \ell_{\text{sm}}^{l}. \tag{10}$$

The joint unsupervised training strategy of multiresolution features makes the spectrum motion features of different levels directly participate in the fitting process of the unsupervised learning framework, which enhances the final feature extraction capability of the SMF-Net.

C. Cross-Domain Feature Extraction for Classification

As mentioned above, our SMF-UL enables us to directly perform the feature learning on an HSI to be classified (in-domain version). This is also a conventional process followed by most of the previous mainstream works. However, the problem brought by this is that the network requires relearning the feature adaptive for the target HSI from scratch when facing a new HSI scene. This is a time-consuming process in practical application, which cannot meet the intelligent requirements of fast processing and analyze for HSIs. Since HSIs all have continuous spectral variation information, for the network, learning the generalized knowledge of perceiving spectral variation can also help to capture the spectral variation information of different HSIs, which provides the possibility for cross-domain spectrum motion feature extraction. Therefore, we propose the strategy of cross-domain spectrum motion feature extraction for our learning framework, as shown in Fig. 4. First, we utilize amounts of unlabeled HSIs to construct the source dataset. The model is pretrained on it to obtain the robust capability of feature extraction with generalization. Then, the trained model is as a feature extractor to directly extract the spectrum motion feature of the target HSI to be classified. Fine-tuning on the target HSI is optional because the improvement of the fine-tuning may be negligible with the size of the source dataset expanding. In this way, we can take the necessary feature learning process ahead on the source dataset of different domains, so the laborious feature learning for the target HSI in the conventional process can be avoided.

D. Source Dataset Expansion for Training

To make the model obtain the ability of cross-domain feature extraction, the model requires to be fully trained on the large-size source dataset to learn the generalized feature, while it is difficult to expand the hyperspectral source dataset in the past because of the inconsistency of sensor, image size, band number, and other technical parameters in different HSI scenes. For our learning framework, the data adaption strategy is essential, i.e., continuous three-channel images are generated as input band by band for one HSI scene. As mentioned above, for an HSI data $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$, we take three consecutive single-band images in HSI as a three-channel image. Therefore, the spectral difference of two adjacent bands $\boldsymbol{B}_i, \boldsymbol{B}_{i+1} \in \mathbb{R}^{H\times W}(i = 1, 2, \ldots, C-2)$ is transformed into the brightness variation information of two three-channel images $\boldsymbol{I}_i(\boldsymbol{B}_{i-1}, \boldsymbol{B}_i, \boldsymbol{B}_{i+1})$ and $\boldsymbol{I}_{i+1}(\boldsymbol{B}_i, \boldsymbol{B}_{i+1}, \boldsymbol{B}_{i+2})$. Thus, $C-2$ three-channel images can be obtained as training samples from an HSI scene with C-bands. Simultaneously, we perform random cropping of the same size when generating the three-channel images from different HSIs so that the samples of different HSIs all meet the input size of the network. In this way, the HSIs with different parameters captured by various sensors can be integrated into a unified source dataset, thus greatly expanding the size of training samples. The rapidly growing mass of unlabeled HSI data can also be effectively utilized at the same time. The expansion of the source dataset enables the network to learn a more robust capability of feature extraction, so enhancing the generalization performance of the feature extractor in the face of target HSI, which will be analyzed in the experimental subsection.

IV. Experimental Results and Analysis

The environmental hardware is an Intel Core i9-7900X processer, 128-GB RAM, and Nvidia Geforce RTX 2080 Ti GPU. Programs of the proposed method are implemented by Python and Pytorch libraries.

A. Datasets

To conduct the experiment and demonstrate the proposed method comprehensively, we select 14 public hyperspectral data: Indian Pines (IP) [65], Salinas (SA), Pavia University (PU), Houston 2013 (HO13) [66], the Kennedy Space Center (KSC), the Pavia Center (PC), Cuprite (CU), Botswana (BO), Dioni (DI), Houston 2018 (HO18) [67], Chikusei (CH), the WHU-Hi dataset [68] that includes Hong Hu (HH), Long Kou (LK), and Han Chuan (HC) three scenes. Their details and parameters are shown in Table II, and their false-color images are shown in Fig. 5. Except that Hyperion is a spaceborne sensor on the E0-1 satellite, other sensors are airborne.

Among them, IP, SA, PU, and HO13 are the target HSI data to be classified, and other HSIs are merely the source data for unsupervised spectrum motion feature learning. Such a scheme makes the target dataset contain both rural (IP and SA) and urban (PU and HO13) scenes to comprehensively verify the performance of different methods. The source dataset also contains both kinds of scenes (rural scenes: CU, BO, DI, CH, HH, LK, and HC; urban scenes: KSC, PC, and HO18), enabling the network to learn a generalized knowledge fully. In subsequent experiments, if the specific composition of source data and target data changes, we will restate it in the corresponding subsection.

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TABLE II
DETAILED INFORMATION OF ALL SELECTED HYPERSPECTRAL DATA

	Data Name	Image Size	Number of Bands	Sensor	Spectral range	Spatial resolution
Target Data	Indian Pines (IP)	145×145	200	AVIRIS	0.40-2.50 μm	20 m
	Salinas (SA)	512×217	204	AVIRIS	0.40-2.50 μm	3.7 m
	Pavia University (PU)	610×340	103	ROSIS	0.43-0.86 μm	1.3 m
	Houston 2013 (HO13)	349×1905	144	ITRES CASI-1500	0.38-1.05 μm	2.5 m
Source Data	Kennedy Space Center (KSC)	512×614	176	AVIRIS	0.40-2.50 μm	18 m
	Pavia Center (PC)	1096×492	102	ROSIS	0.43-0.86 μm	1.3 m
	Cuprite (CU)	614×512	224	AVIRIS	0.37-2.48 μm	20 m
	Botswana (BO)	256×1476	145	Hyperion	0.40-2.50 μm	30 m
	Dioni (DI)	250×1376	176	Hyperion	0.40-2.50 μm	\
	Houston 2018 (HO18)	601×2384	48	ITRES CASI-1500	0.38-1.05 μm	1 m
	Chikusei (CH)	2517×2335	128	Hyperspec-VNIR-C	0.36-1.02 μm	2.5 m
	Hong Hu (HH)	940×475	270	Headwall Nano-Hyperspec	0.40-1.00 μm	0.043 m
	Long Kou (LK)	550×400	270	Headwall Nano-Hyperspec	0.40-1.00 μm	0.463 m
	Han Chuan (HC)	1217×303	274	Headwall Nano-Hyperspec	0.40-1.00 μm	0.109 m

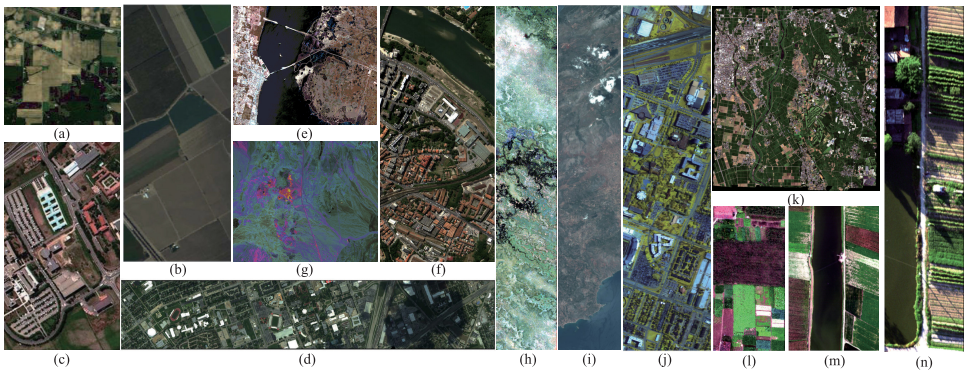


Fig. 5. False-color images of different HSI scenes. (a) IP. (b) SA. (c) PU. (d) HO13. (e) KSC. (f) PC. (g) CU. (h) BO. (i) DI. (j) HO18. (k) CH. (l) HH. (m) LK. (n) HC.

Besides, to verify the effectiveness of the proposed method under the condition of a limited number of samples, 20 labeled samples are selected from each ground category as the training samples for the target dataset when training the classifier (only ten labeled samples are selected for the seventh and ninth categories on the IP scene because of the few sample number), and the rest of all labeled samples are utilized for testing.

B. Implementation Details

The model is trained with the Adam optimizer, the batch size is set to 4, the learning rate is set to 0.0001, the momentum is set to 0.9, and the weight of decaying is set to 0.000001 when training. The weight for smooth regularization ℓ_{sm} in unsupervised loss function is set to $\lambda = 50$. Besides, the cropped size of three-channel images on the source dataset is set to 512 × 192 for all HSI data. When fine-tuning or directly training with the target HSI is required, the cropped size is also set to 512 × 192 for the target HSI, except that the IP with a smaller size (145 × 145) is set for 128 × 128.

As for the SVM classifier with radial basis function (RBF) kernel, a cross-validation strategy is utilized to obtain the optimal regularization parameter λ and kernel parameter φ in the range of $\lambda = \{2^{-2}, 2^{-1}, \ldots, 2^{7}\}$ and $\varphi = \{2^{-2}, 2^{-1}, \ldots, 2^{7}\}$, respectively.

Ten trials are conducted to ensure the reliability of the experiment, and the final results are average values. We randomly select various distributions of training samples at each trial to verify the performance of different approaches. To guarantee the fairness of comparison, we set up the same random number combination for different approaches to ensure that training samples are completely identical at each trial. Besides, the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient κ are taken as the evaluation criterion for evaluating the comprehensive performance. The corresponding full-area classification maps are produced as the qualitative evaluation criteria.

C. Analysis of Source Data Expansion

To first analyze the impact of source data expansion on the generalization performance of the network, we gradually expand the size of the source dataset to observe the performance variation of the SMF-UL on the four different target HSI scenes. We utilize the OA of classification for intuitive evaluation, and the results as shown in Fig. 6. Since a total of ten HSI scenes will be used as the source data, we add them one by one to expand the size of the source dataset. With the expansion of the source dataset size, the performance of the SMF-UL first decreases and then increases on all different target HSI scenes. The reason for the initial decrease in the performance may be that the small-size source dataset constructed by data in different domains interferes with the feature learning of the network to some extent. However, the following increase in the performance demonstrates that the generalization capacity of the SMF-UL can indeed

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TABLE III
RESULTS (OA) OF ABLATION EXPERIMENT ON THE FOUR TARGET HSI DATA (%)

Scheme	Joint Unsupervised Strategy	Source Data Expansion	Fine-Tuning	IP	SA	PU	HO
Baseline				80.61±1.96	95.28±0.78	78.31±3.51	83.90±1.05
A	✓			83.77±1.80	96.31±0.51	84.79±2.42	85.56±1.64
B	✓	✓		**87.94**±2.49	**97.41**±0.67	90.13±2.30	**87.74**±1.71
C	✓	✓	✓	87.22±2.71	95.47±0.86	**91.54**±2.45	85.73±1.92

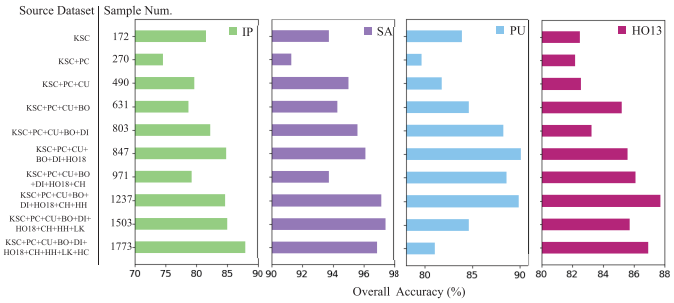


Fig. 6. OA of the SMF-UL varies with the source dataset gradually expanding on the four target HSI scenes.

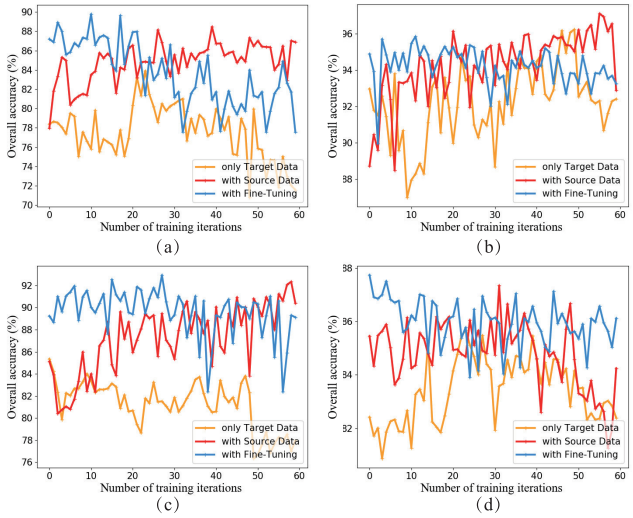


Fig. 7. OA of the SMF-UL varies with the iteration number in the different training modes. (a) IP. (b) SA. (c) PU. (d) HO13.

increase with the expansion of the source dataset, especially for IP, SA, and HO13 target scenes. As for the PU scene, the performance of SMF-UL has an apparent decrease after the LK and HC scenes are added to the source dataset, which is probably because the domain discrepancy between the two newly added rural source scenes (LK and HC) and PU urban target scene is too large. Therefore, in follow-up experiments, we will further explore the impact of domain discrepancy on the generalization capacity of SMF-UL.

D. Analysis of Training Iteration Number

As for the unsupervised spectrum motion feature learning, the number of training iterations is a key parameter, which affects the feature extraction ability of the final model directly. Therefore, to determine a reasonable and universal number of training iterations, we produce a curve graph of OA varying with the iteration number in the different training modes, as shown in Fig. 7. Three curves in different colors represent three training modes. In the first mode, the model is pretrained directly on the target HSI data (i.e., in-domain version.). In the second mode, the model is pretrained on the source data but not fine-tuned. In the third mode, the model pretrained on the source data is further added to the fine-tuning. The source data composition here is consistent with the optimal source data composition corresponding to the four-target HSI in Section IV-C. As we can see, the model performance generally fluctuates and increases with the increase in the pretraining iterations. The performance of the model pretrained with the source data is usually better and more stable than the model pretrained with only the target data. As for the pretrained model added with fine-tuning, the model performance is good at the beginning and then degrades with the increase in the fine-tuning iterations. Therefore, we set the iterations number of the pretraining on the source dataset or the target data to 50, and the iterations number of the fine-tuning on the target HSI data is set to 10. We continue to follow this setting in the subsequent experiments.

E. Ablation Experiment

To verify the respective performance of the joint unsupervised strategy, source data expansion, and fine-tuning, an ablation experiment is well-designed. The results are shown in Table III. The method without any strategies is taken as the baseline method, and other schemes add such three strategies one by one. The baseline and Scheme A directly train the network on the target HSI data. As we can see, the joint unsupervised strategy enables to enhance the classification performance. This denotes that the strategy effectively utilizes the multiresolution outputs to construct additional loss functions to train the network, which improves the final feature extraction capacity of the network. Scheme B further utilizes the strategy of source dataset expansion, and the improvement of classification performance denotes that the SMF-UL acquires a more robust capacity of feature extraction when pretraining on a large-size source dataset. The size of the source dataset here is set according to the optimal data combination in the above subsection. Therefore, relatively sufficient training with large-size datasets may lead to a saturation of performance. Although Scheme C further fine-tunes the network on the target HSI data with a fixed number of training iterations, the fine-tuning fails to result in a significant improvement in network performance, and there is even some performance degradation. Therefore, we consider taking fine-tuning as an

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TABLE IV

ANALYSIS RESULTS OF ACCURACY WITH DIFFERENT IN-DOMAIN APPROACHES ON THE FOUR HSI TARGET SCENES (%) (20 TRAINING SAMPLES IN EACH CATEGORY)

Data	criterion	Unsupervised			Semi-Supervised	Supervised		Unsupervised
		EMPs	LBP	3DCAE	3DAES	SpectralFormer	SSRN	SMF-UL
IP	OA	64.42±0.79	84.17±1.15	66.46±1.89	77.69±1.79	59.56±6.30	83.50±2.54	**87.94**±2.49
	AA	63.51±0.67	83.48±0.84	65.23±1.36	75.48±0.94	59.83±4.94	82.01±2.07	**84.88**±3.55
	κ	61.38±1.08	82.13±1.32	62.40±2.12	74.84±1.95	54.05±6.18	81.36±2.81	**86.38**±2.77
SA	OA	82.86±1.18	88.51±1.57	86.27±1.22	81.20±1.30	84.40±2.01	91.37±1.99	**97.41**±0.67
	AA	86.65±0.54	92.72±1.20	89.71±0.70	89.02±1.10	89.01±1.77	96.12±0.63	**96.25**±0.95
	κ	81.54±1.29	87.25±1.73	84.79±1.32	79.31±1.41	82.22±2.13	90.40±2.20	**97.12**±0.75
PU	OA	75.22±1.75	86.96±1.42	76.50±3.15	80.20±1.96	76.44±2.57	89.22±2.85	**90.13**±2.30
	AA	75.71±0.30	84.14±2.11	74.57±3.74	74.49±1.88	78.17±0.95	87.47±2.92	**87.85**±2.54
	κ	70.75±1.83	86.49±1.74	69.86±3.54	74.75±2.28	68.28±2.46	86.49±3.66	**87.26**±2.86
HO13	OA	80.51±1.56	82.94±1.33	82.36±2.01	83.03±1.31	80.12±2.36	**88.50**±2.07	87.74±1.71
	AA	81.09±0.83	83.14±1.11	82.55±2.32	86.56±1.28	80.10±3.18	**90.21**±1.09	88.35±1.39
	κ	79.79±1.69	81.57±1.43	80.94±2.17	81.65±1.41	77.79±2.51	**87.81**±2.24	86.75±1.85

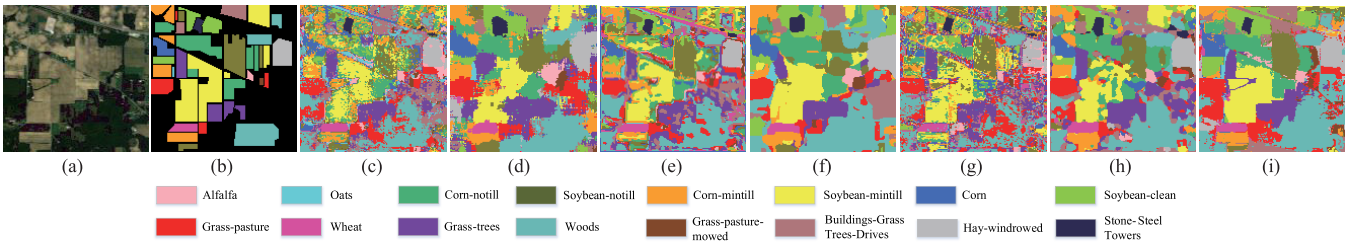


Fig. 8. Full-area classification maps of different in-domain approaches on the IP scene. (a) False-color image. (b) Ground-truth map. (c) EMPs. (d) LBP. (e) 3DCAE. (f) 3DAES. (g) SpectralFormer. (h) SSRN. (i) SMF-UL.

optional step based on requirements in practical applications, which will be further analyzed in the following experiment sections.

F. Experiment for Comparison

1) Compared With In-Domain Methods: Due to the characteristic of the proposed method, except for being purely compared with cross-domain methods, the SMF-UL can also be used as a data extension method to compare with in-domain methods. Therefore, as for in-domain methods (i.e., training and testing on the same HSI scene), we select diverse kinds of methods to comprehensively compare with SMF-UL, including two traditional feature extraction methods, extended morphological profiles (EMPs) [69], local binary patterns (LBP) [9], an unsupervised feature learning method 3-D convolutional autoencoder (3DCAE) [34], a semisupervised feature learning method 3-D-autoencoder with Siamese network (3DAES) [42], and two influential end-to-end supervised classification methods: spectral–spatial residue network (SSRN) [18] and SpectralFormer [15]. The SMF-UL here denotes a feature extraction and classification method enhanced by other unlabeled HSI data (Scheme B), and the composition of the unlabeled HSI data is consistent with the optimal source data in Sections IV-D and IV-E. The comparison results are as shown in Table IV. Furthermore, to make a qualitative comparison intuitively, the corresponding full-area classification maps of different methods are also produced, as shown in Figs. 8–11.

There are two main reasons for the nonhigh accuracy and noisy classification maps. On the one hand, the numbers of training samples that we set are under the limited sample size (20 training samples in each category), which leads to lower accuracy and more noise. On the other hand, the proposed method learns the spectral variation information and takes the enhanced spectral feature as a processing unit of SVM, so the classification errors will lead to scatter noises. However, the proposed method is still the best-performing method among all comparative methods.

As we can see, the traditional feature extraction methods, EMPs and LBP, obtain results competitive to deep-learning-based methods with a limited number of training samples. However, the visual effect of their classification maps is poor because of the numerous scatter noises resulting from the relatively low accuracy of the EMPs, and the wrong areas and boundaries resulting from the excessive texture features extracted by the LBP. The semisupervised feature learning method 3DAES achieves better results than the merely unsupervised feature learning method 3DCAE because of the participation of the labeled samples in the learning process. Although the above four methods all utilize the SVM for classification, the isolated scatter noises are effectively eliminated because 3DCAE and 3DAES capture the spatial information. However, there are still numerous patchy noises. Especially, the 3DAES with higher accuracy also exists the phenomenon of the expansion of ground objects' boundary and distortion of ground objects' distribution. As a supervised end-to-end method, the SSRN obtains higher accuracy (the highest OA in the HO13 scenes) and significantly eliminates patchy noises. Thus, it enables to achieve a better effect of classification no matter with accuracy or visualization. Although the Spectralformer also belongs to the supervised method, it merely takes the spectral sequence as the unit of processing. Besides, the transformer architecture is hard to be adequately trained with the condition of a limited number of

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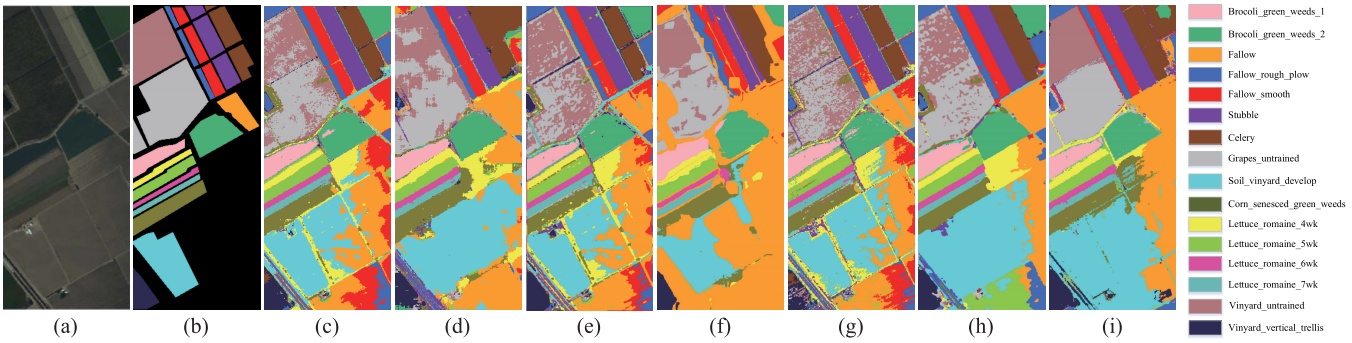


Fig. 9. Full-area classification maps of different in-domain approaches on the SA scene. (a) False-color image. (b) Ground-truth map. (c) EMPs. (d) LBP. (e) 3DCAE. (f) 3DAES. (g) SpectralFormer. (h) SSRN. (i) SMF-UL.

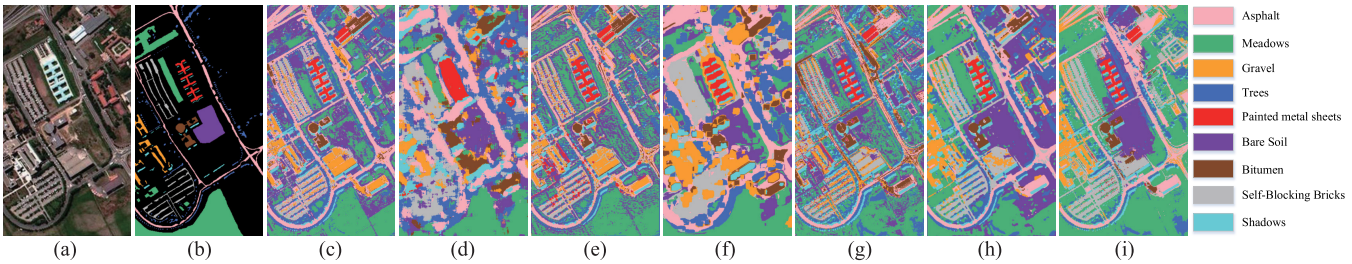


Fig. 10. Full-area classification maps of different in-domain approaches on the PU scene. (a) False-color image. (b) Ground-truth map. (c) EMPs. (d) LBP. (e) 3DCAE. (f) 3DAES. (g) SpectralFormer. (h) SSRN. (i) SMF-UL.

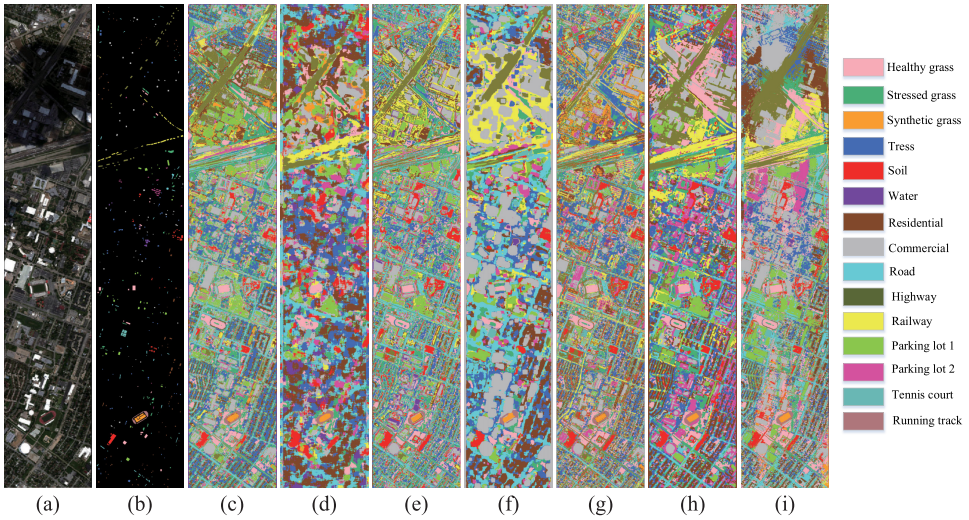


Fig. 11. Full-area classification maps of different in-domain approaches on the HO13 scene. (a) False-color image. (b) Ground-truth map. (c) EMPs. (d) LBP. (e) 3DCAE. (f) 3DAES. (g) SpectralFormer. (h) SSRN. (i) SMF-UL.

training samples. These reasons lead to lower accuracy and numerous scatter noises of Spectralformer. By contrast, our unsupervised feature learning method SMF-UL obtains the highest accuracy on the IP, SA, and PU of three HSI scenes and the closest to the highest accuracy on the HO13 scene. The superiority of SMF-UL is more prominent in the rural scenes (i.e., IP and SA) with a more large range of planar ground objects. Compared with other methods, the SMF-UL indeed gives full play to the advantages of spectral sequences as discriminant features through learning the spectrum motion features. This is reflected in the fewer noises with higher accuracy and the detailed characterization of classification maps. The full-area classification map of the SMF-UL enables to produce a distribution of ground objects closest to the truth of the original image, and some details easily lost with other methods can also be identified and recovered (such as the grass–trees sandwiched between the soybean–mintill on the IP scene, the painted metal sheets, and shadows on the PU scene). All these indicate that the SMF-UL with the robust capability of feature extraction enables to achieve better classification results.

2) Compared With Cross-Domain Methods: To verify the cross-domain performance of the SMF-UL, we further take a comparison with several cross-domain methods. At present, there are few pure cross-domain related works. The comparison requires that the match between source data and

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TABLE V

ANALYSIS RESULTS OF ACCURACY WITH DIFFERENT CROSS-DOMAIN APPROACHES ON THE THREE HSI TARGET SCENES (%) (20 TRAINING SAMPLES IN EACH CATEGORY)

Data	criterion	Supervised				Unsupervised		
		DFSL	RN-FSC	CDPM	Gia-CFSL	UM^{2}L	SMF-UL	SMF-UL (ft)
IP	OA	75.72±2.08	66.32±1.27	69.89±1.76	80.35±0.97	**83.78**±1.57	76.56±3.01	81.54±1.41
	AA	71.87±1.75	63.80±2.20	78.52±1.51	80.11±2.72	**81.77**±0.79	76.32±1.94	79.72±1.90
	κ	72.74±2.29	62.24±1.45	65.49±1.92	77.89±1.10	**80.30**±1.75	73.65±3.35	80.11±1.57
SA	OA	89.93±0.54	87.88±1.39	89.15±1.57	92.30±0.76	94.35±1.22	94.58±0.76	**95.84**±1.26
	AA	92.01±0.74	90.67±1.01	93.37±0.71	95.36±0.33	95.06±0.67	94.75±0.52	**95.27**±1.64
	κ	88.83±0.59	86.57±1.53	88.27±1.73	91.44±0.84	94.94±1.35	93.98±0.85	**95.37**±1.39
PU	OA	85.77±4.62	84.24±3.25	82.83±2.99	86.10±2.92	90.41±2.66	88.29±3.14	**91.84**±2.01
	AA	85.94±2.58	83.59±3.18	84.47±1.38	84.84±2.33	91.06±1.53	85.81±3.28	**91.22**±2.33
	κ	81.77±5.51	79.66±3.89	76.76±3.55	82.17±3.46	88.41±3.36	84.95±3.85	**89.43**±2.52

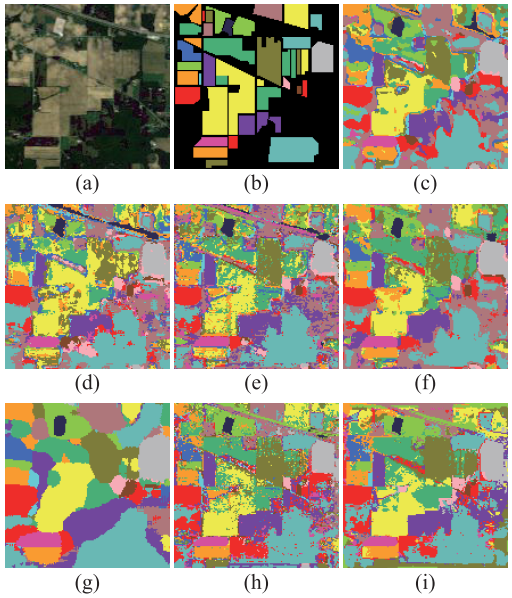


Fig. 12. Full-area classification maps of different cross-domain approaches on the IP scene. (a) False-color image. (b) Ground-truth map. (c) DFSL. (d) RN-FSC. (e) CDPM. (f) Gia-CFSL. (g) UM^{2}L. (h) SMF-UL. (i) SMF-UL (ft).

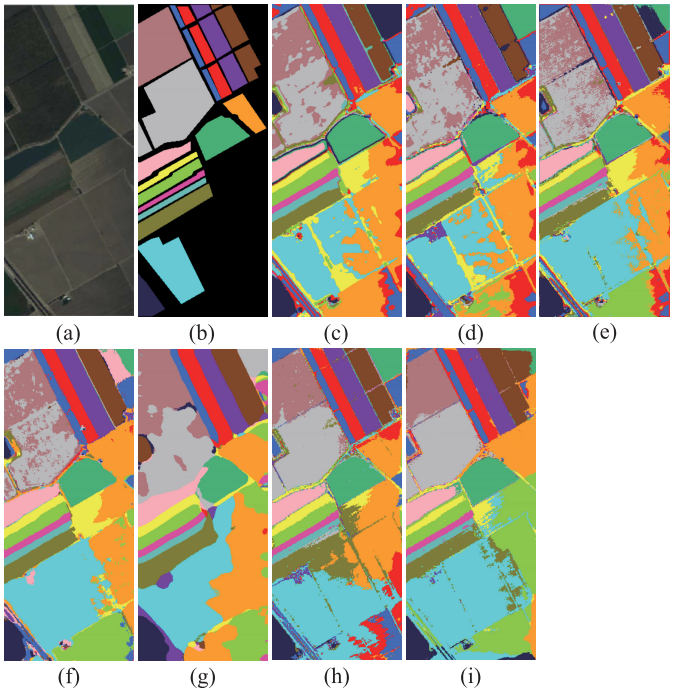


Fig. 13. Full-area classification maps of different cross-domain approaches on the SA scene. (a) False-color image. (b) Ground-truth map. (c) DFSL. (d) RN-FSC. (e) CDPM. (f) Gia-CFSL. (g) UM^{2}L. (h) SMF-UL. (i) SMF-UL (ft).

target data is completely consistent to ensure fairness, so we select three typical pure cross-domain methods for comparison: including deep few-shot learning (DFSL) [49], relation network of few-shot learning (RN-FSC) [50], unsupervised metalearning with multiview constraints (UM^{2}L) [37], and cross-domain pretrained model (CDPM) [55]. Besides, we further select the latest and state-of-the-art method to demonstrate the superiority of the proposed method: graph information aggregation cross-domain few-shot learning (Gia-CFSL) [57]. Among them, DFSL, RN-FSC, CDPM, and Gia-CFSL all require labeled samples with source data at the pretraining phase (except for those categories with fewer than 200 labeled samples and 200 training samples in each category), while UM^{2}L and our SMF-UL do not require. All detailed settings of such five cross-domain methods refer to the original literature. Thus, as for the cross-domain comparison, HO13, KSC, BO, and CH are selected to compose the source dataset, so the HO13 is removed from the target dataset. The results are shown in Table V, and the corresponding full-area classification maps are produced for a qualitative comparison intuitively, as shown in Figs. 12–14.

As for two supervised cross-domain methods, DFSL has a better performance than RN-FSC on all three target HSI data, which denotes that the deep residual network of DFSL obtains a more robust capacity of deep feature extraction than the relation network of RN-FSC by the same supervised feature learning. The more discriminative deep feature improves the classification performance of DFSL with a limited number of target labels, and the patchy noises of classification maps are decreased compared with RN-FSC. As for the cross-domain pretrained model, CDPM can simultaneously utilize multiple source data to pretrain the model with data-specific layers and task-specific layers. By loading the pretrained weight of feature analysis layers, CDPM improves the accuracy and convergence speed on the target domain, but it does not show a better performance than other few-shot learning methods under the condition of limited-size samples. Even so, to solve the small samples problem of the target domain, the procedure of the pretraining is still necessary for the deep

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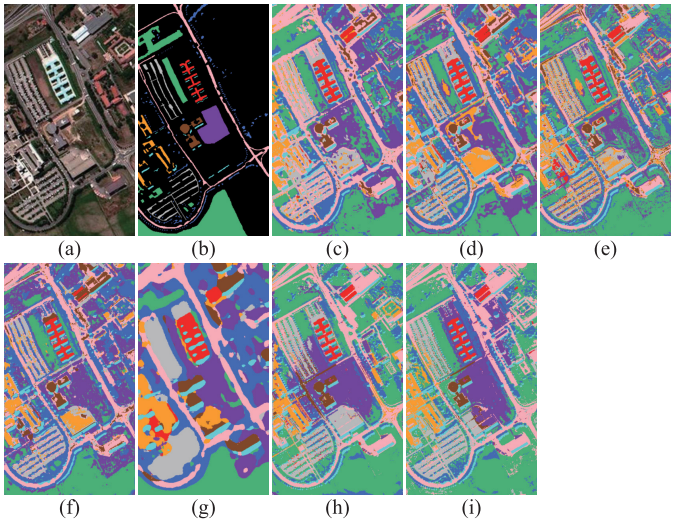


Fig. 14. Full-area classification maps of different cross-domain approaches on the PU scene. (a) False-color image. (b) Ground-truth map. (c) DFSL. (d) RN-FSC. (e) CDPM. (f) Gia-CFSL. (g) UM^{2}L. (h) SMF-UL. (i) SMF-UL (ft).

model of CDPM (13 layers of convolution); otherwise, limited-size samples are easier to lead to the overfitting of deep model and worse performance. Besides, the entire network of CDPM composed of 1×1 convolution kernels also leads to numerous scatter noises on the classification maps. As for another cross-domain learning method, Gia-CFSL executes joint training with abundant labels in the source domain and few labels in the target domain, improving the accuracy of target data. Benefiting from the advanced domain alignment strategy, Gia-CFSL achieves the best performance in the supervised cross-domain methods and further decreases the patchy noises of classification maps, obtaining relatively smooth visual effects. Although two unsupervised cross-domain methods are pretrained without any labels, they obtain an apparent improvement in performance on the target HSI data. Among them, UM^{2}L constructs unsupervised metalearning tasks with multiview constraints to effectively dig out the features on numerous unlabeled samples, while RN-FSC, DFSL, CDPM, and Gia-CFSL merely utilize a limited number of labeled samples in this phase. Hence, the deeper relation network of UM^{2}L is capable of training adequately. Therefore, UM^{2}L achieves superior results and also surpasses the version that is no fine-tuning of SMF-UL (except for the OA on the SA), obtaining the highest accuracy on the IP. However, although the higher accuracy of UM^{2}L results in effectively eliminating the noises of classification maps, the phenomenon of the expansion of ground objects' boundary and distortion of ground objects' distribution is relatively prominent because its processing unit with a large-size patch (28×28) results in an excessive smoothing of spatial information.

As for the nonhigh accuracy and noisy classification maps of SMF-UL, there are two main reasons. On the one hand, the number of training samples that we set is under the limited sample size (20 training samples in each category), which leads to relatively low accuracy compared with the performance under enough labels. On the other hand, the proposed method learns the spectral variation information and takes the enhanced spectral feature as a processing unit of SVM, so the classification errors will lead to scatter noises. However, the proposed method still has the best performance among all comparative methods. The version of SMF-UL that is no fine-tuning still achieves better performance than all supervised methods (except for Gia-CFSL on the IP). Besides, the defect in classification maps of UM^{2}L can be avoided effectively by SMF-UL because classification maps of SMF-UL well preserve the distribution and boundary of ground objects consistent with the original image. The SMF-UL added with fine-tuning further improves the performance, and it achieves the highest accuracy on the SA and PU. Therefore, on the basis of the SMF-UL, the classification maps of SMF-UL (ft) further reduce scatter noises and present a better visual effect. By contrast, the classification maps of other methods under limited sample size present relatively worse visual effects because of the numerous patchy noises and distortion of ground objects.

The pure cross-domain comparison with a consistent match between source data and target data proves the superiority of the SMF-UL as the cross-domain method and its robust capacity of feature extraction with generalization for a new target HSI data.

G. Analysis of Domain Discrepancy

To analyze the impact of domain discrepancy of the source data and target data on the generalization capacity of SMF-UL, we will conduct experiments in two parts: sensor discrepancy and scene discrepancy.

1) Sensor Discrepancy: The type of sensor will directly determine the attributes of the HSI data, so we first select source data based on sensor type. Besides, to further control variables, we keep the scenes of source and target HSI data similar during selection. The results are shown in Table VI. As we can see, the sensor consistency between the source data and the target data fails to result in a significant improvement in classification performance compared to the sensor inconsistency. Therefore, the sensor discrepancy is not the main factor affecting the generalization ability of SMF-UL, and the impact resulting from the HSI data of different sensors is obviously weakened because of the data composition of three-channel images and the normalization of spectral values.

2) Scene Discrepancy: The type of scene will directly determine the categories of the ground objects. For instance, urban scenes usually are more complicated and contain more linear objects, but rural scenes are usually simpler and contain a larger range of planar objects. Therefore, we second select source data based on scene type. Besides, we keep each sensor different in the source data of the same scene during selection for the fairness of comparison. The results are shown in Table VII. As we can see, the scene consistency between source data and target data enables to result in a significant improvement in classification performance compared to the scene inconsistency. It denotes that the SMF-UL trained on the source dataset constituted by one kind of scene has a better generalization ability of feature extraction on the target HSI with similar scenes. Therefore, the SMF-UL is more suitable for training on a large-size source dataset composed of similar

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TABLE VI
ACCURACY WITH THE SOURCE DATASET CONSTITUTED BY HSI DATA OF DIFFERENT SENSORS (%) (20 TRAINING SAMPLES IN EACH CATEGORY)

Source \ Target	IP (AVIRIS)	SA (AVIRIS)
CU (AVIRIS)	77.68±2.11	92.08±1.43
BO (Hyperion)	78.11±1.75	91.30±1.40
LK (Headwall Nano-Hyperspec)	78.66±0.98	94.68±1.47
Source \ Target	**PU (ROSIS)**	**HO13 (ITRES CASI-1500)**
PC (ROSIS)	82.43±1.95	84.66±1.71
HO18 (ITRES CASI-1500)	82.38±3.70	82.36±1.81

TABLE VII
ACCURACY WITH THE SOURCE DATASET CONSTITUTED BY HSI DATA OF DIFFERENT SCENES (%) (20 TRAINING SAMPLES IN EACH CATEGORY)

Source \ Target	IP (Rural)	SA (Rural)	PU (Urban)	HO13 (Urban)
CU+BO+LK (Rural)	85.02±1.97	93.82±1.38	79.04±5.01	84.09±1.46
KSC+PC+HO18 (Urban)	78.99±2.30	93.35±1.35	86.04±1.90	86.50±1.61

HSI scenes and being applied to feature extraction on the target HSI data with the corresponding similar scenes.

H. Analysis of Running Time

To further illustrate the time-efficient advantage of the SMF-UL when processing a new HSI target data to be classified, the running-time results of different deep-learning-based approaches for comparison are given in Table VIII. As for most feature extraction methods, the time consumption on pretraining and feature extraction is a major proportion, but the time with training and classification of the classifier (SVM) is generally negligible. While the end-to-end methods integrate the feature learning into the training of a network for classification, the time with training and classification is longer. As for the in-domain comparison, the SMF-UL directly trained on the target HSI data (Scheme A) is superior to the two feature extraction methods in accuracy and efficiency, which denotes the superiority of the in-domain version of SMF-UL. On the other hand, more importantly, the SMF-UL enables us to take the feature learning process ahead by being pretrained on a large-size source dataset. In this way, the SMF-UL enables to directly extract features from the new target HSI data (Scheme B), which effectively shortens the processing time in practice without any retraining, while other pure in-domain methods require laborious retraining from scratch when facing new HSI data to be classified. Besides, the expansion of the source dataset also enhances the generalization capacity of the SMF-UL. Fine-tuning is optional in practice, which will bring about extra time consumption.

As for the cross-domain methods pretrained on a completely identical source dataset (HO13, KSC, BO, and CH), the time for pretraining of the SMF-UL is slightly longer than DFSL and CDPM but significantly less than two metalearning-based methods RN-FSC and UM^{2}L. The SMF-UL takes a shorter time than other methods for fine-tuning and feature extraction, which also denotes its advantage for fast responding to a new target HSI data, especially for the version without fine-tuning. CDPM and RN-FSC and UM^{2}L are end-to-end classification frameworks with a similar schedule of training and testing, but CDPM apparently has higher efficiency on pretraining, fine-tuning, and classification because it takes the global image as input, while RN-FSC and UM^{2}L both take the data cubes as input. Although Gia-CFSL utilizes the knowledge of the source domain when training, it still requires retraining for a new HSI, and the time consumption is huge. SMF-UL added with fine-tuning shows an obvious improvement in performance, which is due to the source dataset here being smaller than the one mentioned above for comparison of the ablation experiment. Model performance unsaturation due to a smaller source dataset provides ample room for performance improvement with fine-tuning. Therefore, fine-tuning can be selected in practice according to the comprehensive requirement of accuracy, timeliness, and the size of the source dataset.

I. Analysis of Spectrum Motion Feature

To further analyze the spectrum motion feature deeply, we take the PU target data with 103 original bands as an example to discuss its properties from different perspectives in this section. First, several three-channel images composed of consecutive three bands in the HSI data and their corresponding spectrum motion features extracted by the SMF-UL are selected for visual presentation, as shown in Fig. 15. As we can see, the spectral variation information for the HSI data can be reflected in the brightness change of different three-channel images with a static scene. The SMF-UL enables us to perceive such spectral variation through learning the brightness change of three-channel images produced by HSI data, so achieving unsupervised spectrum motion feature learning. Visualization of the spectrum motion feature shows that the ground objects with the same category have instantaneous similarity in spectral changes, thus presenting similar colors in the image. Such motion feature is a conducive supplement for the original spectral feature, which significantly enhances the discrimination of spectral features.

To further illustrate it visually, second, we produce three groups of curves of spectral value change, as shown in Fig. 16. The first set of curves denotes the final feature for classification, which is obtained by the concatenation between the original spectral feature and its motion feature extracted by SMF-UL. From the curve of the spectrum motion feature, we can roughly distinguish the different spectral variations of different ground objects, which makes it easier to distinguish the ground objects with a similar original spectrum. Because the spectrum motion feature corresponding to each band $\boldsymbol{F} = \{\boldsymbol{u}, \boldsymbol{v}\} \in \mathbb{R}^{H\times W\times 2}$ contains a horizontal component $\boldsymbol{u}$ and

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TABLE VIII

RUNNING TIME OF DIFFERENT APPROACHES ON THE FOUR TARGET HSI SCENES (20 TRAINING SAMPLES IN EACH CATEGORY)

Target HSI	Criterion	In-Domain							Cross-Domain						
		3DCAE	3DAES	Spectral-Former	SSRN	SMF-UL A	SMF-UL B	SMF-UL C	DFSL	RN-FSC	CDPM	Gia-CFSL	UM^{2}L	SMF-UL	SMF-UL (ft)
IP	labeled samples	300	300	300	300	300	0+300	0+300	11000+300	11000+300	11000+300	11000+300	0+300	0+300	0+300
	pre-training (min)	59.52	6.61	\	\	10.49	147.03	147.03	23.93	320.43	60.27	\	317.04	65.61	65.61
	fine-tuning(s)	\	822.26	\	\	\	\	114.21	\	995.11	30.63	\	95.88	\	114.21
	feature extraction(s)	31.04	93.41	\	\	13.82	13.82	13.82	9.56	\	\	\	\	13.82	13.82
	training(s)	1.52	2.11	160.13	86.85	1.24	0.97	1.04	0.84	\	\	6161.12	\	1.28	0.99
	classification(s)	1.11	1.57	4.41	4.97	0.71	0.68	0.59	0.44	12.59	0.12	3.82	8.76	0.91	0.73
	OA (%)	66.46	77.69	59.56	83.50	83.77	87.94	87.22	75.72	66.32	69.89	80.35	83.78	76.56	80.54
SA	labeled samples	320	320	320	320	320	0+320	0+320	11000+320	11000+320	11000+320	11000+320	0+320	0+320	0+320
	pre-training (min)	314.92	21.23	\	\	14.35	129.52	129.52	23.93	320.43	60.27	\	317.04	65.61	65.61
	fine-tuning(s)	\	849.98	\	\	\	\	146.03	\	878.20	166.53	\	155.86	\	146.03
	feature extraction(s)	118.77	278.19	\	\	14.41	14.41	14.41	34.31	\	\	\	\	14.41	14.41
	training(s)	1.22	2.01	128.41	108.29	0.99	1.26	1.03	0.69	\	\	5674.49	\	0.89	0.94
	classification(s)	3.28	2.57	18.79	32.45	2.99	2.62	2.89	1.52	64.39	0.67	18.67	77.21	2.78	2.40
	OA (%)	86.27	81.20	84.40	91.37	96.31	97.41	95.47	89.93	87.88	89.15	92.30	94.35	94.58	95.84
PU	labeled samples	180	180	180	180	180	0+180	0+180	11000+180	11000+180	11000+180	11000+180	0+180	0+180	0+180
	pre-training (min)	509.64	6.32	\	\	6.23	92.97	92.97	23.93	320.43	60.27	\	317.04	65.61	65.61
	fine-tuning(s)	\	343.81	\	\	\	\	78.39	\	171.81	196.61	\	128.93	\	78.39
	feature extraction(s)	211.36	203.51	\	\	9.92	9.92	9.92	80.03	\	\	\	\	9.92	9.92
	training(s)	0.72	0.49	90.60	48.83	0.49	0.64	0.56	0.47	\	\	3485.06	\	0.85	0.48
	classification(s)	2.66	1.57	13.3	20.34	0.86	1.03	0.92	0.94	76.52	0.82	27.29	122.72	0.99	0.85
	OA (%)	76.5	80.20	76.44	89.22	84.79	90.13	91.54	85.77	84.24	82.83	86.10	90.41	88.29	91.84
HO13	labeled samples	300	300	300	300	300	0+300	0+300							
	pre-training (min)	1548.91	29.52	\	\	11.75	115.83	115.83							
	fine-tuning(s)	\	73.98	\	\	\	\	137.38							
	feature extraction(s)	598.84	1566.78	\	\	41.58	41.58	41.58							
	training(s)	1.78	1.28	86.45	117.71	1.04	0.88	0.96							
	classification(s)	2.01	1.73	20.09	76.64	0.78	0.82	0.94							
	OA (%)	82.36	83.03	80.12	88.50	85.56	87.74	85.73							

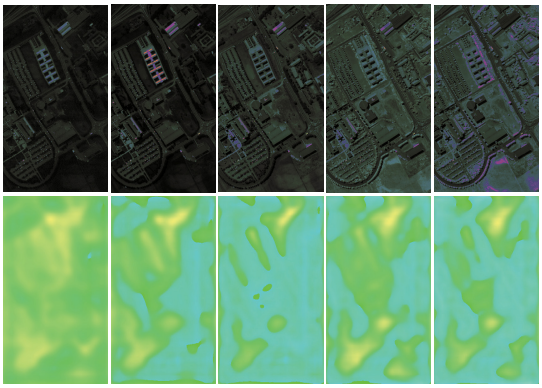


Fig. 15. Visualization of three-channel images (first row) and the corresponding spectrum motion features (second row).

a vertical component $\boldsymbol{v}$, we show two groups of components separately to obtain the second and third set of curves, which will show the movement of the spectrum more clearly. The separate display of the two groups of components $\boldsymbol{u}$ and $\boldsymbol{v}$ allows us to observe more clearly that the difference in spectral variation enhances the discrimination of the original spectrum. Some indistinguishable ground objects with a similar original spectrum become distinguishable due to the addition of spectral variation information, such as the asphalt and gravel objects.

Then, we utilize the T-SNE dimension-reduction visualization approach to further present distinctly such advantage of spectrum motion feature for identification and classification, as shown in Fig. 17. The features corresponding to 600 labeled samples in each category in PU data are selected for display.

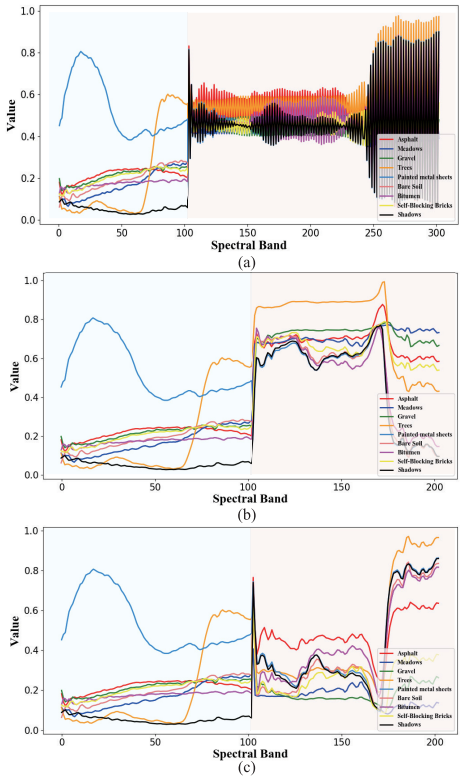


Fig. 16. Three groups of curves of spectral value change. (a) Original spectrum and spectrum motion feature. (b) Original spectrum and the horizontal component $\boldsymbol{u}$ of spectrum motion feature. (c) Original spectrum and the vertical component $\boldsymbol{v}$ of spectrum motion feature.

As for the original feature, different ground object samples are close to each other even overlapping in the feature space,

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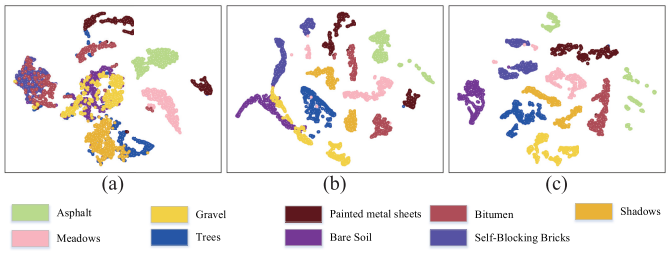


Fig. 17. T-SNE visualization results on the PU scene. (a) Original feature. (b) Spectrum motion feature extracted by the in-domain version of SMF-UL. (c) Spectrum motion feature extracted by the data-expansion version of SMF-UL.

so it is difficult to identify and distinguish. The SMF-UL merely trained on the target data effectively mitigates such phenomenon and makes the most samples easier for classification. However, there are still a small number of samples of different categories close to each other in the feature space. The SMF-UL trained on the source data with expansion further leads to aggregation for the samples of the same class and separation for the samples of different classes in the feature space, which significantly reduces the difficulty of identification and classification. It also demonstrates again that the SMF-UL acquires a more robust capability of feature extraction with generalization when pretraining on a large-size source dataset.

V. Conclusion

In recent years, deep-learning-based HSI classification methods have achieved significant development. However, the general process followed by most of the previous mainstream works retrains the network from scratch to obtain the capability of feature extraction adaptive to the target image when facing a new HSI to be classified. It is a laborious process that cannot reasonably meet the requirements of rapid processing and analysis for HSI in practice.

Aiming at this problem, we consider taking the process of feature learning ahead and making the network have a robust capability of feature extraction with generalization through pretraining. On account of the spectral variation information generally existing in the HSIs, we rethink the HSI data from a perspective of spectral sequence and construct an SMF-UL to perceive such variation. To make the network learn more generalized knowledge on a large-size dataset, we develop an extendable training dataset construction method, which can integrate arbitrary HSI data with different specification parameters into a unified training dataset. Besides, it also enables the rapidly growing mass of unlabeled HSI data to be effectively utilized.

Extensive experiments show that, except for the competitive in-domain version of SMF-UL, the SMF-UL acquires a more robust capability of feature extraction with generalization when pretraining on a large-size source dataset. The comparison results of in-domain and cross-domain experiments denote that the SMF-UL gives full play to the advantages of spectral sequences as discriminant features through learning the spectrum motion features, which are reflected in the fewer noises with higher accuracy, and the detailed characterization of full-area classification maps. Then, the experiment about the impact of domain discrepancy shows that the SMF-UL is more suitable for training a large-size source dataset composed of similar HSI scenes and being applied to the target HSI data with the corresponding similar scenes.

Therefore, the SMF-UL enables us to take ahead the feature learning by being pretrained on a large-size source dataset. In this way, the SMF-UL with generalization enables fast response to a new target HSI data without retraining, which effectively shortens the processing time in practice. Fine-tuning is optional in practice according to the comprehensive requirement of accuracy, timeliness, and the size of the source dataset.

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References

[1] Y. Zhong et al., "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.
[2] L. Zhang, L. Zhang, D. Tao, and X. Huang, "On combining multiple features for hyperspectral remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 879–893, Mar. 2012.
[3] B. Liu, X. Yu, P. Zhang, A. Yu, Q. Fu, and X. Wei, "Supervised deep feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1909–1921, Apr. 2018.
[4] B. Rasti et al., "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, 2020.
[5] A. Agarwal, T. El-Ghazawi, H. El-Askary, and J. Le-Moigne, "Efficient hierarchical-PCA dimension reduction for hyperspectral imagery," in *Proc. IEEE Int. Symp. Signal Process. Inf. Technol.*, Dec. 2007, pp. 353–356.
[6] J. Li and Y. Qian, "Dimension reduction of hyperspectral images with sparse linear discriminant analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2011, pp. 2927–2930.
[7] J. Xia, M. D. Mura, J. Chanussot, P. Du, and X. He, "Random subspace ensembles for hyperspectral image classification with extended morphological attribute profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 4768–4786, Sep. 2015.
[8] S. Jia, J. Hu, Y. Xie, L. Shen, X. Jia, and Q. Li, "Gabor cube selection based multitask joint sparse representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3174–3187, Jun. 2016.
[9] W. Li, C. Chen, H. Su, and Q. Du, "Local binary patterns and extreme learning machine for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3681–3693, Jul. 2015.
[10] S. Jia, J. Hu, J. Zhu, X. Jia, and Q. Li, "Three-dimensional local binary patterns for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2399–2413, Apr. 2017.
[11] L. Zhang, Q. Zhang, B. Du, X. Huang, Y. Y. Tang, and D. Tao, "Simultaneous spectral-spatial feature selection and extraction for hyperspectral images," *IEEE Trans. Cybern.*, vol. 48, no. 1, pp. 16–28, Jan. 2018.
[12] C. Xing, L. Ma, and X. Yang, "Stacked denoise autoencoder based feature extraction and classification for hyperspectral images," *J. Sensors*, vol. 2016, pp. 1–10, Jun. 2016.
[13] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.
[14] A. Ma et al., "Fast sequential feature extraction for recurrent neural network-based hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5920–5937, Sep. 2021.
[15] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

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[16] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral-spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.

[17] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[18] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Aug. 2017.

[19] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral-spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, May 2020.

[20] M. E. Paoletti, S. Moreno-Álvarez, and J. M. Haut, "Multiple attention-guided capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.

[21] T. Song, Y. Wang, C. Gao, H. Chen, and J. Li, "MSLAN: A two-branch multidirectional Spectral–Spatial LSTM attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[22] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[23] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.

[24] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[25] Z. Xue, M. Zhang, Y. Liu, and P. Du, "Attention-based second-order pooling network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9600–9615, Nov. 2021.

[26] Y. Shen et al., "Efficient deep learning of nonlocal features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 1–15, Jul. 2020.

[27] Z. Zheng, Y. Zhong, A. Ma, and L. Zhang, "FPGA: Fast patch-free global learning framework for fully end-to-end hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5612–5626, Aug. 2020.

[28] D. Wang, B. Du, and L. Zhang, "Fully contextual network for hyperspectral scene parsing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[29] G. Jiang, Y. Sun, and B. Liu, "A fully convolutional network with channel and spatial attention for hyperspectral image classification," *Remote Sens. Lett.*, vol. 12, no. 12, pp. 1238–1249, Dec. 2021, doi: 10.1080/2150704X.2021.1978582.

[30] Y. Sun, B. Liu, X. Yu, A. Yu, Z. Xue, and K. Gao, "Resolution reconstruction classification: Fully octave convolution network with pyramid attention mechanism for hyperspectral image classification," *Int. J. Remote Sens.*, vol. 43, no. 6, pp. 2076–2105, Mar. 2022, doi: 10.1080/01431161.2022.2054299.

[31] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, Nov. 2020.

[32] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[33] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[34] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Apr. 2019.

[35] S. Zhang, M. Xu, J. Zhou, and S. Jia, "Unsupervised spatial–spectral CNN-based feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[36] W. Wei, S. Xu, L. Zhang, J. Zhang, and Y. Zhang, "Boosting hyperspectral image classification with unsupervised feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[37] K. Gao, B. Liu, X. Yu, and A. Yu, "Unsupervised meta learning with multiview constraints for hyperspectral image small sample set classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3449–3462, 2022.

[38] B. Liu, A. Yu, X. Yu, R. Wang, K. Gao, and W. Guo, "Deep multiview learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7758–7772, Sep. 2021.

[39] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[40] M. Zhu, J. Fan, Q. Yang, and T. Chen, "SC-EADNet: A self-supervised contrastive efficient asymmetric dilated network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[41] B. Fang, Y. Li, H. Zhang, and J. C.-W. Chan, "Collaborative learning of lightweight convolutional neural network and deep clustering for hyperspectral image semi-supervised classification with limited training samples," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 164–178, Mar. 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271620300125

[42] S. Jia et al., "A semisupervised Siamese network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[43] D. Gadot and L. Wolf, "PatchBatch: A batch augmented loss for optical flow," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4236–4245.

[44] A. Dosovitskiy et al., "FlowNet: Learning optical flow with convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 2758–2766.

[45] E. Ilg, N. Mayer, T. Saikia, M. Keuper, A. Dosovitskiy, and T. Brox, "FlowNet 2.0: Evolution of optical flow estimation with deep networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1647–1655.

[46] Z. Ren, J. Yan, B. Ni, B. Liu, X. Yang, and H. Zha, "Unsupervised deep learning for optical flow estimation," in *Proc. 31st AAAI Conf. Artif. Intell.*, S. Singh and S. Markovitch, Eds. San Francisco, CA, USA: AAAI Press, 2017, pp. 1495–1501. [Online]. Available: http://aaai.org/ocs/index.php/AAAI/AAAI17/paper/view/14388

[47] J. J. Yu, A. W. Harley, and K. G. Derpanis, "Back to basics: Unsupervised learning of optical flow via brightness constancy and motion smoothness," 2016, *arXiv:1608.05842*.

[48] L. Liu et al., "Learning by analogy: Reliable supervision from transformations for unsupervised optical flow estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6488–6497.

[49] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[50] K. Gao, B. Liu, X. Yu, J. Qin, P. Zhang, and X. Tan, "Deep relation network for hyperspectral image few-shot classification," *Remote Sens.*, vol. 12, no. 6, p. 923, Mar. 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/6/923

[51] K. Gao, W. Guo, X. Yu, B. Liu, A. Yu, and X. Wei, "Deep induction network for small samples classification of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3462–3477, 2020.

[52] C. Zhong, J. Zhang, S. Wu, and Y. Zhang, "Cross-scene deep transfer learning with spectral feature adaptation for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2861–2873, 2020.

[53] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[54] X. He, Y. Chen, and P. Ghamisi, "Heterogeneous transfer learning for hyperspectral image classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3246–3263, May 2020.

[55] H. Lee, S. Eum, and H. Kwon, "Exploring cross-domain pretrained model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[56] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[57] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 30, 2022, doi: 10.1109/TNNLS.2022.3185795.

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[58] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.
[59] R. Hang, F. Zhou, Q. Liu, and P. Ghamisi, "Classification of hyperspectral images via multitask generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1424–1436, Feb. 2021.
[60] J. Wang, S. Guo, R. Huang, L. Li, X. Zhang, and L. Jiao, "Dual-channel capsule generation adversarial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.
[61] H. Wu and S. Prasad, "Semi-supervised deep learning using pseudo labels for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1259–1270, Mar. 2018.
[62] Y. Zhang, X. Wang, X. Jiang, and Y. Zhou, "Marginalized graph self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.
[63] Y. Zhang, J. Wu, C. Zhou, Z. Cai, J. Yang, and P. S. Yu, "Multi-view fusion with extreme learning machine for clustering," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 5, pp. 1–23, Sep. 2019, doi: 10.1145/3340268.
[64] D. Sun, X. Yang, M. Y. Liu, and J. Kautz, "PWC-Net: CNNs for optical flow using pyramid, warping, and cost volume," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 8934–8943.
[65] M. F. Baumgardner, L. L. Biehl, and D. A. Landgrebe. (Sep. 2015). *220 Band Aviris Hyperspectral Image Data Set: June 12, 1992 Indian Pine Test Site 3*. [Online]. Available: https://purr.purdue.edu/publications/1947/1
[66] N. Acito, S. Matteoli, A. Rossi, M. Diani, and G. Corsini, "Hyperspectral airborne 'Viareggio 2013 trial' data collection for detection algorithm assessment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2365–2376, Mar. 2016.
[67] Y. Xu et al., "Advanced multi-sensor optical remote sensing for urban land use and land cover classification: Outcome of the 2018 IEEE GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1709–1724, May 2019.
[68] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425720303825
[69] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2008.



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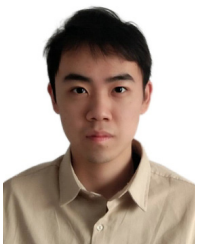
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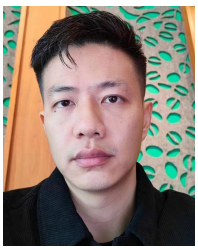
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